

Coordinator & Worker Swarm Execution

How Complex High-Level Tasks Are Broken Down and Executed in Parallel

Dynamic Hierarchical Task Decomposition

When a user assigns a complex project, the Coordinator uses a structured planning prompt to analyze dependencies, formulate execution branches, and launch parallel sub-agents via the `spawn_agent` tool.

Sample Task Decomposition Tree

Root Coordinator: "Comprehensive Due Diligence on European AI FinTechs"

Branch 1 (Registry Agent):
Scrapes Companies House for corporate registration & filings.

Branch 2 (OSINT Agent):
Harvests C-level leadership, LinkedIn profiles, and verified work emails.

Branch 3 (Tech Stack Agent):
Fingerprints homepage HTML for 40+ technology signatures.

RECURSIVE DEPTH LIMITS & GUARDRAILS

To prevent runaway sub-agent spawning loops, Fathom enforces strict depth limits (default max depth: 2). Coordinators can spawn workers, but workers cannot spawn infinite child trees without explicit permission.

TOKIO JOINSET MULTI-THREADING

All spawned workers run as concurrent tasks inside a Tokio JoinSet. If one branch encounters a slow network request or rate limit, sibling branches continue executing at full speed.

Fault-Tolerant Merging: If one sub-agent fails due to an anti-bot block, the coordinator gracefully incorporates partial findings from other branches without failing the overall run.